



VIT[®]

Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

Lab assessment 1

Class No: CH2025260502854

Course code: BMAT202P

Slot: L5+L6

Total Marks: 24

Course Name: Probability and Statistics

Faculty Name: Dr. Basua Debananda

Answer all the questions using R

1. Create a data frame in R with the following columns: **Name, Emp.ID, Age, Gender, Food Preference (Veg/Non-Veg), and Salary.**
 - a. Generate the **Gender** and **Food Preference** variables using the factor () function.
 - b. Display the **summary statistics** of the complete dataset.
 - c. Extract and display data **based on Gender**.
 - d. Extract and display data **based on Food Preference**.
 - e. Visualize the **Salary distribution** using a **Boxplot**, grouped by:
 - i. Gender
 - ii. Food Preference
2. Consider the following data:
`name <= c("A", "B", "C")` marks <= `c(65, 78, 85)`
Create a data frame named result and write R commands to:
 - a) Display only the marks column.
 - b) Find the maximum mark. (R code and output)
 - c) Write R commands to import a CSV file named student data.csv and display the first three rows.
 - d) Display the number of rows and columns.
3. **Using the ToothGrowth Dataset:** The dataset ToothGrowth in R contains measurements of tooth length (len) in guinea pigs given different doses of Vitamin C (dose) through two delivery methods (supp: OJ and VC)
 - a) Load the dataset and display its structure.
 - b) Draw a boxplot to compare tooth length (len) across supplement types (supp).
 - c) Draw a grouped boxplot to study the effect of both dose (dose) and supplement type (supp) on tooth length.
 - d) Draw a bar plot showing the average tooth length for each dose level.
 - e) Interpret the graphical results.



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Lab Assessment 2

Class No: CH2025260502854

Course code: BMAT202P

Slot: L5+L6

Total Marks: 16

Course Name: Probability and Statistics

Faculty Name: Dr. Basua Debananda

Answer all the questions using R

1. Import the built-in **airquality** dataset in R and compute the following:
 - a) Identify the key **numerical attributes** (Ozone, Solar.R, Wind, Temp).
 - b) Compute **measures of central tendency** and **Dispersion**.
 - c) Interpret what these measures indicate about **air quality trends**, such as typical ozone levels, common temperature ranges, and overall atmospheric conditions.

Collect at least 60 students and analyse the data by using descriptive statistics and interpret the results.

2. Outlier Impact Study

Given:

`x1 <- c(10, 12, 15, 18, 20, 22, 25, 30)` `x2 <- c(10, 12, 15, 18, 20, 22, 25, 300)`

Write an R program to:

- a) Compute mean, median, and standard deviation for both datasets.
- b) Compute coefficient of variation for both.
- c) Compute skewness and kurtosis for both datasets.
- d) Analyze the effect of the extreme value.
- e) Identify which statistical measures are robust.